KAIZEN MATHS

Lesson starter

Further Vectors: Lines, Planes and Products

Planes and Cross Products

Name

S1

Find each cross product.

Find $a \times b$ for $a = \begin{pmatrix} 6 \\ 4 \\ -6 \end{pmatrix}$ and $b = \begin{pmatrix} 4 \\ 4 \\ -4 \end{pmatrix}$.

S2

Find the Cartesian equation of each plane.

A plane has normal vector $n = \begin{pmatrix} -3 \\ -5 \\ 4 \end{pmatrix}$ and passes through P = (-5, 2, 2). Find its Cartesian equation.

S3

Find the equation of the plane through each set of three points.

Find the equation of the plane through A = (-6, 1, 6), B = (-12, 0, 1), and C = (0, -5, 1).

S4

Find the point where each line meets the plane.

Find where the line $r = \begin{pmatrix} -5 \\ 1 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} 3 \\ 3 \\ -2 \end{pmatrix}$ meets the plane $-5x - 6y + 4z = -100$.

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KAIZEN MATHS

Exit ticket

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Name _____

EA1

Find each cross product.

Find a × b for a = $\begin{pmatrix} 6 \\ -1 \\ -4 \end{pmatrix}$ and b = $\begin{pmatrix} -5 \\ -2 \\ 1 \end{pmatrix}$.

EA2

Find the Cartesian equation of each plane.

A plane has normal vector n = $\begin{pmatrix} -4 \\ -6 \\ 2 \end{pmatrix}$ and passes through P = (-5, 2, -5). Find its Cartesian equation.

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Exit ticket

Further Vectors: Lines, Planes and Products

Planes and Cross Products

Name _____

EB1

Find the equation of the plane through each set of three points.

Find the equation of the plane through A = (-4, -2, -5), B = (-1, 4, -6), and C = (2, 1, -10).

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KAIZEN MATHS

Exit ticket

Further Vectors: Lines, Planes and Products

Planes and Cross Products

Name _____

EA1

Find each cross product.

Find a × b for a = $\begin{pmatrix} 6 \\ -1 \\ -4 \end{pmatrix}$ and b = $\begin{pmatrix} -5 \\ -2 \\ 1 \end{pmatrix}$.

EA2

Find the Cartesian equation of each plane.

A plane has normal vector n = $\begin{pmatrix} -4 \\ -6 \\ 2 \end{pmatrix}$ and passes through P = (-5, 2, -5). Find its Cartesian equation.

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Further Vectors: Lines, Planes and Products

Planes and Cross Products

Name _____

EB1

Find the equation of the plane through each set of three points.

Find the equation of the plane through A = (-4, -2, -5), B = (-1, 4, -6), and C = (2, 1, -10).

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Show your working

Teacher answers & working

Further Vectors: Lines, Planes and Products · Planes and Cross Products

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S1

Find each cross product.

Find a × b for a = $\begin{pmatrix} 6 \\ 4 \\ -6 \end{pmatrix}$ and b = $\begin{pmatrix} 4 \\ 4 \\ -4 \end{pmatrix}$.

Answer:

$\mathrm{a} \times \mathrm{b} = \begin{pmatrix} 8 \\ 0 \\ 8 \end{pmatrix}$

Working:

Use $\mathrm{a} \times \mathrm{b} = \begin{pmatrix} a_2b_3 - a_3b_2 \\ a_3b_1 - a_1b_3 \\ a_1b_2 - a_2b_1 \end{pmatrix}$.

$\mathrm{a} \times \mathrm{b} = \begin{pmatrix} 4(-4) - -6(4) \\ -6(4) - 6(-4) \\ 6(4) - 4(4) \end{pmatrix}$

$\mathrm{a} \times \mathrm{b} = \begin{pmatrix} 8 \\ 0 \\ 8 \end{pmatrix}$

S2

Find the Cartesian equation of each plane.

A plane has normal vector n = $\begin{pmatrix} -3 \\ -5 \\ 4 \end{pmatrix}$ and passes through P = (-5, 2, 2). Find its Cartesian equation.

Answer:

$-3\mathrm{x} - 5\mathrm{y} + 4\mathrm{z} = 13$

Working:

Use $\mathrm{n} \cdot \mathrm{r} = \mathrm{n} \cdot \mathrm{P}$.

$$\begin{pmatrix} -3 \\ -5 \\ 4 \end{pmatrix} \cdot \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} -3 \\ -5 \\ 4 \end{pmatrix} \cdot \begin{pmatrix} -5 \\ 2 \\ 2 \end{pmatrix}$$

$\mathrm{n} \cdot \mathrm{P} = 13$

$-3\mathrm{x} - 5\mathrm{y} + 4\mathrm{z} = 13$

Teacher answers & working

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S3

Find the equation of the plane through each set of three points.

Find the equation of the plane through A = (-6, 1, 6), B = (-12, 0, 1), and C = (0, -5, 1).

Answer:

$-25x - 60y + 42z = 342$

Working:

Find two direction vectors in the plane.

$$\mathrm{AB} = \begin{pmatrix} -12 \\ 0 \\ 1 \end{pmatrix} - \begin{pmatrix} -6 \\ 1 \\ 6 \end{pmatrix} = \begin{pmatrix} -6 \\ -1 \\ -5 \end{pmatrix}$$

$$\mathrm{AC} = \begin{pmatrix} 0 \\ -5 \\ 1 \end{pmatrix} - \begin{pmatrix} -6 \\ 1 \\ 6 \end{pmatrix} = \begin{pmatrix} 6 \\ -6 \\ -5 \end{pmatrix}$$

A normal vector is AB × AC.

$$\mathrm{n} = \begin{pmatrix} -6 \\ -1 \\ -5 \end{pmatrix} \times \begin{pmatrix} 6 \\ -6 \\ -5 \end{pmatrix} = \begin{pmatrix} -25 \\ -60 \\ 42 \end{pmatrix}$$

Use $\mathrm{n} \cdot \mathrm{r} = \mathrm{n} \cdot \mathrm{A}$.

$-25x - 60y + 42z = 342$

S4

Find the point where each line meets the plane.

Find where the line $\mathrm{r} = \begin{pmatrix} -5 \\ 1 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} 3 \\ 3 \\ -2 \end{pmatrix}$ meets the plane $-5x - 6y + 4z = -100$.

Answer:

Intersection = (4, 10, -5)

Working:

Substitute the line into the plane equation.

$$\mathrm{r} = \begin{pmatrix} -5 \\ 1 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} 3 \\ 3 \\ -2 \end{pmatrix} = \begin{pmatrix} -5 + 3\lambda \\ 1 + 3\lambda \\ 1 - 2\lambda \end{pmatrix}$$

$-5(-5 + 3\lambda) - 6(1 + 3\lambda) + 4(1 - 2\lambda) = -100$

Solving gives $\lambda = 3$.

Substitute $\lambda = 3$ into the line.

Intersection = (4, 10, -5)

Teacher answers & working

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EA1

Find each cross product.

Find a × b for a = $\begin{pmatrix} 6 \\ -1 \\ -4 \end{pmatrix}$ and b = $\begin{pmatrix} -5 \\ -2 \\ 1 \end{pmatrix}$.

Answer:

$a \times b = \begin{pmatrix} -9 \\ 14 \\ -17 \end{pmatrix}$

Working:

Use $a \times b = \begin{pmatrix} a_2b_3 - a_3b_2 \\ a_3b_1 - a_1b_3 \\ a_1b_2 - a_2b_1 \end{pmatrix}$.

$a \times b = \begin{pmatrix} -1(1) - -4(-2) \\ -4(-5) - 6(1) \\ 6(-2) - -1(-5) \end{pmatrix}$

$a \times b = \begin{pmatrix} -9 \\ 14 \\ -17 \end{pmatrix}$

EA2

Find the Cartesian equation of each plane.

A plane has normal vector n = $\begin{pmatrix} -4 \\ -6 \\ 2 \end{pmatrix}$ and passes through P = (-5, 2, -5). Find its Cartesian equation.

Answer:

$-4x - 6y + 2z = -2$

Working:

Use $n \cdot r = n \cdot P$.

$\begin{pmatrix} -4 \\ -6 \\ 2 \end{pmatrix} \cdot \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} -4 \\ -6 \\ 2 \end{pmatrix} \cdot \begin{pmatrix} -5 \\ 2 \\ -5 \end{pmatrix}$

$n \cdot P = -2$

$-4x - 6y + 2z = -2$

Teacher answers & working

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EB1

Find the equation of the plane through each set of three points.

Find the equation of the plane through A = (-4, -2, -5), B = (-1, 4, -6), and C = (2, 1, -10).

Answer:

$-27x + 9y - 27z = 225$

Working:

Find two direction vectors in the plane.

$$AB = \begin{pmatrix} -1 \\ 4 \\ -6 \end{pmatrix} - \begin{pmatrix} -4 \\ -2 \\ -5 \end{pmatrix} = \begin{pmatrix} 3 \\ 6 \\ -1 \end{pmatrix}$$

$$AC = \begin{pmatrix} 2 \\ 1 \\ -10 \end{pmatrix} - \begin{pmatrix} -4 \\ -2 \\ -5 \end{pmatrix} = \begin{pmatrix} 6 \\ 3 \\ -5 \end{pmatrix}$$

A normal vector is AB × AC.

$$n = \begin{pmatrix} 3 \\ 6 \\ -1 \end{pmatrix} \times \begin{pmatrix} 6 \\ 3 \\ -5 \end{pmatrix} = \begin{pmatrix} -27 \\ 9 \\ -27 \end{pmatrix}$$

Use $n \cdot r = n \cdot A$.

$-27x + 9y - 27z = 225$